Rethinking Image-Scaling Attacks

When Scaling Meets Black-box ML: A Cloud API Misclassifies

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Concrete scenario

- Users upload HR photos to a cloud Image Analysis API. The service downsizes images before classification.
- Attackers craft HR inputs whose perturbations survive preprocessing and scaling, causing label flips.

Problem

- ML systems include preprocessing h and scaling g before the model f.
- Vulnerabilities in g and f can compound in decision-based black-box settings.

Setting

- Labels only (no scores), unknown internals; $g \circ h$ can be inferred once via brute force.
- Goal: fewer queries, lower scaled L2, and higher ASR by exploiting scaling.

● HR (pipeline-aware) ● LR (model-only)

Method: Scaling-aware Noise Sampling (SNS)

Sample in L, project to H; plug-and-play for decision-based attacks

- Attacks iterate via boundary/direction search, noise sampling, and update.
- SNS samples noise in LR space L and projects back to HR space H to align with $g \circ h$.
- This strategy guides search along directions that survive preprocessing and scaling, with perturbations hidden.
- Efficient projection uses an imprecise preimage sufficient for gradient estimation.

The method integrates with HSJ and Sign-OPT, as demonstrated in prior work.

SNS Illustration

Sample in L → project to H → perturbations persist through h and g

The attacker targets $F = f \circ g \circ h$

The approach also includes offline EOT to handle randomized preprocessing without extra queries.

Defenses and Key Insight

Preprocessing and detection; end-to-end implications

Preprocessing defenses (Quiring et al.)

- Median filtering is robust to outliers but can slow convergence; improved gradient estimation mitigates this effect.
- Randomized filtering selects a random pixel in each window.

Detection defenses (Kim et al.)

- Unscaling and min-filtering apply thresholded distortion after transforms.
- Centered spectrum examines high-frequency peaks.

- Four of five state-of-the-art defenses are reported circumvented under a decision-based threat model.
- Randomized filtering is characterized as uniformizing scaling in expectation in the literature, leaving no exploitable subspace.

Defense Illustration

A histogram after unscaling shows overlap between benign and HR adversarial examples; randomized filtering induces uniform scaling in expectation.

Core Results

HR (SNS) vs LR across datasets and an API

- HR attacks reduce scaled L2 and increase ASR versus LR at equal queries (ImageNet, CelebA) under no defenses.
- With median filtering and improved gradient estimation, HR attacks continue to outperform LR in scaled L2 and ASR.
- Under randomized filtering, HR attacks do not surpass LR; this aligns with white-box PGD findings.
- On a cloud API (Tencent), HR HSJ achieves lower scaled L2 and higher ASR at 1–3K queries.

Results Figure

A comparison of HR with SNS to LR baselines shows lower scaled L2 and higher ASR; a median-defense comparison appears as an inset.

● HR with SNS ● LR baseline

An ablation disabling SNS yields HR behavior similar to LR; efficient projection performs comparably to precise projection with far less cost.

Takeaway

End-to-end robustness requires scaling-aware design

- SNS improves efficiency and stealth by making decision-based attacks scaling-aware.
- Most scaling defenses fail end-to-end; randomized filtering addresses the underlying weakness.
- Evaluations are conducted on the full pipeline $F = f \circ g \circ h$ to avoid a false sense of security.

End-to-end aware methods   Model-only defenses are insufficient